

Helix OTA “ Spec-vs-Implementation Alignment Decision Memo (operational / api / v1)

Field	Value
Revision	4
Last modified	2026-06-10T00:00:00Z
Created	2026-06-08
Status	active (decision memo “ additive + group breaking WIDENs landed; row-4 per-device telemetry filters + row-7 group/members pagination RESOLVED 2026-06-10; row-4 richer numeric fields still ingest-gated) Row 4 filters “ per-device telemetry now accepts the spec filter params (50ef5c6, feat(api): per-device telemetry filters + group/members pagination; OpenAPI synced + redocly-clean). Row 7 pagination “ group + members list now paginated. Both previously-DEFERRED WIDEN bits are RESOLVED. Dashboard API client updated in lockstep (b0b8ee2). Still open on row 4: the richer per-item numeric fields duration_ms/ bytes_transferred, blocked on UNVERIFIED ingest (Â§9). WIDEN impl to the fuller spec for the remaining breaking divergences. Executing in sub-batches. Landed: group row 6 id-key id“group_id + row 8 member-add “ batch {device_ids} “ 200 {added, already_member, not_found} (device-registration-aware); row 1 audit actor “ object {user_id, subject}; row 4 structural “ per-device telemetry events“items + next_cursor, newest-first, ?limit/?cursor pagination (handler-side over bounded history). GET /groups/{id}/members “ items[] of {device_id, added_at} (store gained an added_at membership column + ListGroupMembersDetailed on memory + pgx, real-DB parity). Remaining: row 4 DEFERRED bits (richer per-item fields duration_ms/bytes_transferred “ UNVERIFIED ingest; ?event_type/?since/?until/?deployment_id filters),
Landed (2026-06-10)	
Operator ruling (2026-06-08)	

Field

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Landed (2026-06-08, operator-approved additive WIDENs)

Status summary

Authority

Anti-bluff

Owner

Related

row 7 (group + members list pagination “ deferred, groups bounded).

Row 2 (audit query filters) “ ?since/? until RFC3339 bounds on GET /audit (028e656). **Row 5 core (telemetry overview)** “ failure_rate = failure/ (success+failure) + by_state (028e656). **Row 6 partial (member_count)** “ added to GroupView on get/list/update (live count via ListGroupMembers). All three additive (new optional params / new JSON fields), wire-compatible, real-DB parity where a store change applied. Remaining rows (1 audit actor object, 4 per-device telemetry pagination/eventsâ†’items, 8 batch member-add, plus the idâ†’group_id rename) are BREAKING wire changes still gated behind an operator WIDEN/TRIM ruling. A per-divergence decision memo over the eight spec-vs-implementation gaps recorded in [implemented_endpoints.md](#) Â§10 for the audit-read, telemetry-read, and device-group operational surfaces. For each gap it recommends **WIDEN-impl** (bring the handler up to the fuller [operational_endpoints.md](#) spec) or **TRIM-spec** (accept the leaner as-built shape as canonical and amend the prose spec), with rationale, effort, risk, and the blast radius (tests / e2e / HelixQA bank). It is a MEMO only “ it recommends; it changes no server / code, no go.mod, no tests, no existing file.

Recommendations are advisory. The operator decides WIDEN-vs-TRIM per row (Constitution Â§11.4.66 “ closed-set decisions belong to the operator; Â§11.4.122 “ no silent removal of a shipped capability). Per Constitution Â§11.4.6 (no-guessing): every “impl does” cell was confirmed by direct read of the cited handler function in server / internal / api /; every “spec wants” cell was confirmed by direct read of [operational_endpoints.md](#). The Â§10 register at the foot of this memo lists what is **UNVERIFIED** (effort/risk estimates, NFR numbers, store-method availability) and is not asserted as fact.

Helix OTA control-plane / API team
[implemented_endpoints.md](#) (as-built Â§10 gap list “ source of this memo);
[operational_endpoints.md](#) (fuller

Field

Value

prose spec); [endpoints.md](#) (conventions/error-model/RBAC/pagination);
[openapi.yaml](#)

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1. purpose_and_scope

`implemented_endpoints.md` §10 records eight concrete shape divergences between the prose operational spec (`operational_endpoints.md`) and the code that actually shipped (`handlers_audit.go`, `handlers_telemetry.go`, `handlers_group.go`). The as-built doc states the **code is authoritative** but does not decide which side *should* win going forward. This memo makes that decision per gap:

- **WIDEN-impl** “the spec’s fuller shape carries real operational value; raise the handler to it (the as-built doc becomes a transient under-build state, the prose spec stays the target).
- **TRIM-spec** “the leaner as-built shape is sufficient for the 1.0.0-MVP; adopt it as canonical and amend `operational_endpoints.md` + `openapi.yaml` to match (no code churn; doc-sync only).

Scope is exactly the eight §10 gaps across audit / telemetry / groups. Out of scope: rollout (`implemented_endpoints.md` §10 marks it *consistent* with the brick), roles (also *consistent*), and the PLANNED recall endpoint (§8 “a missing feature, not a shape divergence).

2. how_to_read_this_memo

- `impl` does `cites file:func` “the exact handler confirmed by reading the source this pass.
- **WIDEN/TRIM** is the recommendation; the operator owns the final call (§11.4.66).
- **Effort** is a relative T-shirt size (S / M / L) for the handler+store+test work; the absolute hours are UNVERIFIED (§9) because the pgx store-method availability for the wider shapes is not confirmed in this memo.
- **Risk** is the chance the change breaks an existing green test or an as-built consumer.
- **Blast radius** names the test layers and HelixQA bank entries that would change. The HelixQA bank path (`tools/helixqa/banks/atmosphere.yaml`) is the constitutional convention (§11.4.58); whether that exact file exists in this repo is UNVERIFIED (§9) “treat it as “the project’s Challenge bank, wherever it lives”.

3. evidence_basis_confirmed_against_code

Each divergence below was re-confirmed against source this pass (not taken on trust from Â§10):

- **Audit actor flat-string** “handlers_telemetry.go aside, the audit entry is built in handlers_audit.go:auditMiddleware (store.AuditEntry{ActorSubject: claims.Subject, ...}, L36“45) and rendered by toAuditLogEntry into AuditLogEntry; the wire actor is a single string, never a {user_id, subject} object. Spec Â§4.3 shows "actor": { "user_id": ..., "subject": ... } (L234).
- **Audit query params** “handlers_audit.go:handleListAudit reads only action, resource_type, cursor, limit (L133“147). Spec Â§4.3 (L224“226) lists additionally ?actor, ?resource_id, ?since, ?until.
- **Audit response type name** “code emits AuditLogList (handleListAudit, L157); spec Â§4.3 calls it AuditList (L227).
- **Telemetry per-device** “handlers_telemetry.go:handleDeviceTelemetry consumes no query params, returns full history in store order under events (L51“67); per-event TelemetryEventView (L14“22) has no id/success/duration_ms/bytes_transferred. Spec Â§5.1 (L270“292) wants ?limit/?cursor/?event_type/?since/?until/?deployment_id, newest-first, an items array, and those richer fields.
- **Telemetry overview** “handlers_telemetry.go:handleTelemetryOverview (L71“82) returns the minimal TelemetryOverview{EventCounts, Total} (L32“35), no scoping. Spec Â§5.2 (L308“342) wants scope, devices_total, devices_reporting, by_state (latest-per-device), failure_rate, generated_at, plus ?deployment_id/?os/?since/?until.
- **Group view fields + id key** “handlers_group.go:GroupView (L28“33) is {id, name, description, created_at}; no filter_criteria, no member_count, id key is id. Spec Â§6.1 (L39“44) shows group_id, filter_criteria, member_count.
- **Group list pagination** “handlers_group.go:handleListGroup (L79“90) returns all groups, GroupList{Items} (L36“38), no ?name/?limit/?cursor, no next_cursor. Spec Â§6.2 (L53“54) wants paginated {items, next_cursor} + ?name= prefix.
- **Group member add semantics** “handlers_group.go:handleAddGroupMember (L152“171) takes a single MemberAdd{DeviceID} (L41“43) and returns 204. Spec Â§6.4 (L89“98) wants a batch {device_ids: [...]} returning 200 with GroupMemberAddResult{added, already_member, not_found}.
- **Group member list shape** “handlers_group.go:handleListGroupMembers (L139“150) returns GroupMembers{group_id, device_ids:[string]}. Spec Â§6.4 (L76“85) wants paginated items of {device_id, added_at}.

4. decision_table

# divergence	spec wants	impl does (file:func)
1 audit actor shape	nested {user_id, subject}	flat string = subject“” (handlers_audit.toAuditLogEntry)

#	divergence	spec wants	impl does (file:func)
2	audit query filters	?actor,?resource_id,?since,?until (+ existing)	only ?action,?resource_id (handlers_audit_log_list)
3	audit response type name	AuditList	AuditLogList (handlers_audit_log_list)
4	per-device telemetry: pagination + newest-first + filters + richer fields	?limit/?cursor/?event_type/?since/?until/?deployment_id,newest-first, items[] with id/success/duration_ms/bytes_transferred	newest-first items[] params RESOLVED 2 still UNVERIFIED-ingress (handlers_telemetry_overview)
5	telemetry overview: richer fields + scoping	scope,devices_total,devices_reporting,by_state (latest-per-device),failure_rate,generated_at + ? deployment_id/?os/?since/?until	minimal {event_count,failure_rate} + ? (handlers_telemetry_overview) TelemetryOverviewResponse
6		group_id,filter_criteria,member_count	

divergence

spec wants

impl does (file:func)

GroupView
fields + id key

id, no filter_cri
(handlers_group.

GET /groups
+ GET /
7 groups/{id}/
members
pagination

paginated {items, next_cursor} (+ ?name= prefix on
groups)

RESOLVED 2026-06-
now paginated (handl
handleListGroup

group member
8 add: single+204
vs
batch+200+result

batch {device_ids:[...]} â†’ 200 {added,
already_member, not_found}

single {device_id}
(handlers_group.
MemberAdd)

5. per_divergence_rationale

5.1 audit (rows 1â€“3)

The audit subsystemâ€™s only purpose is forensic reconstruction. Two of its three gaps directly defeat that purpose:

- **Row 1 (WIDEN).** The schema keeps a nullable `user_id` precisely so a record outlives the actor (`operational_endpoints.md` Â§4 intro, `ON DELETE SET NULL`). The handler currently flattens to a single subject string. Surfacing `{user_id, subject}` restores the durable join key for cross-actor queries; effort is small because both values already exist at write time (`claims.Subject` is captured; the `users.id` resolution the spec describes in Â§4.1 L140 is the only new lookup). **Effort Sâ€“M. Risk LOW** â€” additive JSON field; existing string consumers break only if they assumed `actor` is a string (the as-built doc says it is, so an as-built consumer would break â€” gate this behind the operatorâ€™s WIDEN decision).
- **Row 2 (WIDEN).** `?since/?until` time-window scoping is the single most-used audit query in incident response; `?actor/?resource_id` complete the â€œwho touched whatâ€” triad. The store seam (`store.AuditFilter`) is already the extension point â€” adding fields there + parsing+validating the RFC-3339 params in `handleListAudit` is the bulk of the work. **Effort M. Risk LOW-MED** â€” new validation paths (malformed `since/until` â†’ 400) must be tested to avoid Â§11.4.1 FAIL-bluffs.
- **Row 3 (TRIM).** Type *name* only; the wire contract (`items, next_cursor`) is identical. Renaming `AuditLogList` â†’ `AuditList` is pure churn. Adopt `AuditLogList` as canonical; fix the prose + openapi. **Effort S (doc only). Risk NONE.**

5.2 telemetry (rows 4â€“5)

- **Row 4 (WIDEN, phased).** Per-device history is unbounded and the handler returns *all of it in insertion order*. At any real device age this is a latency/payload problem and gives no â€œmost-recent-firstâ€” affordance. Pagination + newest-first + `?event_type/?since/?until` filtering are load-bearing and should be widened. The richer fields `split_id` and `success` are cheap derivations; `duration_ms/bytes_transferred` require the telemetry ingest to *carry* those values â€” whether `store.TelemetryRecord` / the ota-protocol event already has them is **UNVERIFIED (Â§9)**. Phase: ship pagination+order+filters first; add the rich numeric fields only once the ingest is confirmed to supply them (else they would be a Â§11.4-bluff of always-zero fields). **Effort Mâ€“L. Risk MED** â€” touches the store query contract (`TelemetryForDevice` gains filter/cursor params) and changes the response *key* (`events` â†’ `items`), which is a breaking wire change an as-built consumer would feel; gate behind the operator decision.
- **Row 5 (WIDEN core / TRIM ?os).** `failure_rate + by_state` computed as *latest-event-per-device* is the honest fleet-health number; the current raw `event_counts` over-weights chatty devices and has no health ratio at all. Widen to `by_state/failure_rate/devices_total/devices_reporting/generated_at`. The latest-per-device aggregate is a `DISTINCT ON (device_id)` query (spec Â§5.3 alludes to it) â€” real work in the store. `?os` scoping is the weakest item (needs a device-OS join, low MVP demand) â€” TRIM it for 1.0.0, keep `?deployment_id + time window`. **Effort L. Risk MED** â€” additive response fields are safe; the new store aggregate must be tested for the zero-devices edge (`failure_rate = 0` when `devices_reporting = 0`, per spec Â§5.2 L345).

5.3 groups (rows 6â€“8)

- **Row 6 (split).** Three sub-decisions: **id key** id (TRIM â€” matches the rest of the API, renaming to group_id churns every group test for cosmetics); **member_count** (WIDEN â€” a cheap COUNT, high-value in list/detail UIs); **filter_criteria** (TRIM for MVP â€” the spec *itself* flags dynamic-membership evaluation as UNVERIFIED and MVP as static-only, Â§6 intro L13â€“14; storing+returning an un-evaluated selector is a latent Â§11.4-bluff â€œfeature that doesnâ€™t workâ€, so defer until the evaluator lands). **Effort Sâ€“M. Risk LOW.**
- **Row 7 (TRIM, defer).** Groups are operator-curated and bounded (not per-device rows), so an un-paginated list is acceptable at MVP scale. Adopt the lean shape as canonical now; document the paginated {items, next_cursor} + ?name= as a future widen triggered if groups ever get created programmatically. **Effort S (doc). Risk LOW.**
- **Row 8 (WIDEN).** Membership is inherently bulk (build a cohort = add many devices). Single-add forces N round-trips + N audit rows and gives no partial-success feedback; the specâ€™s {added, already_member, not_found} is precisely the report an operator needs, and the store method is plural by design (AddGroupMembers). The 204â†’200+body and device_idâ†’device_ids changes are breaking for an as-built single-add consumer â€” gate behind the operator decision. **Effort M. Risk MED** â€” must test idempotent re-add (counted, not error) and unknown-device (not_found, partial success) to avoid a Â§11.4.1 FAIL-bluff on the partial path.

6. overall_recommendation_first_things_first

Order by (operational-harm-if-left-lean) Æ— (low effort/risk first), and honor Â§11.4.72 only where applicable (no audio surface here, so pure operational priority):

1. **Audit query filters (row 2) â€” FIRST.** Highest harm: without ?since/?until/?actor/?resource_id the audit trail cannot answer the incident question it exists for. Medium effort, low-med risk, purely additive query params (no breaking wire change). Biggest forensic value per unit work.
2. **Telemetry overview core fields (row 5, minus ?os) â€” SECOND.** failure_rate + by_state is the fleet-health number operators watch during a rollout; additive response fields are non-breaking. Pairs naturally with the rollout/telemetry story.
3. **Per-device telemetry pagination + newest-first + filters (row 4, fields phase-2) â€” THIRD.** Prevents the unbounded-history payload cliff; defer duration_ms/bytes_transferred until the ingest is confirmed to supply them. Breaking key change (eventsâ†’items) means it needs the operator decision + a consumer sweep, so it sits behind the additive wins.
4. **Group member batch-add (row 8) â€” FOURTH.** Real ergonomics win for cohort building; breaking shape change so it rides with row 4 under the operator decision.
5. **Audit actor object (row 1) â€” FIFTH.** Restores the durable user_id join key; additive but needs the users resolution lookup; lower urgency than time-window filtering.
6. **member_count on GroupView (row 6 partial) â€” SIXTH.** Cheap UI affordance; do it alongside any group-handler touch.
7. **TRIM-spec items (rows 3, 7, and the id-key + filter_criteria + ?os sub-parts of 5/6) â€” ANYTIME, doc-only.** No code; amend operational_endpoints.md + openapi.yaml so the prose stops disagreeing with the shipped code (Â§11.4.6 â€” divergences must not sit silently). Land these as a single doc-sync change because they cost nothing and immediately stop the spec from lying.

Net: do the three TRIM doc-syncs immediately (zero code, removes the standing divergence), then work the WIDEN list in the 1â†’6 order above behind a single operator WIDEN/TRIM

ruling per breaking row (Â§11.4.66). Rows 1, 2, 5 are additive/non-breaking and can land without consumer coordination; rows 4, 8 (and the actor-object visibility change in 1) are breaking wire changes that need the as-built consumer sweep first.

7. blast_radius_summary

layer	rows that touch it	what changes
wire-struct unit tests	1,3,4,5,6,8	JSON tag / field-set assertions for AuditLogEntry, TelemetryEventView, TelemetryOverview, GroupView, MemberAddâ†’batch, AuditLogList/AuditList name
handler unit/integration tests	1,2,4,5,8	query-param parsing + validation (400 on bad since/until/limit), newest-first ordering, batch idempotency + not_found partial success, aggregate edge (failure_rate at 0 devices)
store seam tests	1,2,4,5,8	AuditFilter (actor/resource/time), TelemetryForDevice (filter+cursor), new latest-per-device overview aggregate, AddGroupMemberâ†’AddGroupMembers, member_count query
e2e / HelixQA Challenge bank	1,2,4,5,6,8	audit-filter, fleet-health, device-history, group-CRUD, group-membership Challenges each re-assert the new shape with captured evidence (Â§11.4.69 / Â§11.4.83)
openapi.yaml	all 8	path/schema definitions re-aligned to whichever side wins each row
audit middleware	8	none â€” deriveAuditAction already emits group_member verbs for the members sub-routes (handlers_audit.go), so batch-add stays correctly audited

8. doc_sync_actions_if_trim_chosen

For every row the operator rules TRIM-spec, the prose spec and openapi MUST be amended in the same change so they stop disagreeing with the shipped code (Constitution Â§11.4.6 â€” no silent divergence; Â§11.4.65 â€” keep .md/.html/.pdf siblings in sync):

- Row 3 â†’ operational_endpoints.md Â§4.3: rename response model AuditList â†’ AuditLogList.
- Row 6 (id) â†’ Â§6.1/Â§6.2: change group_id â†’ id in the Group/GroupView examples.
- Row 6 (filter_criteria) â†’ Â§6.1/Â§6.3: mark filter_criteria as â€œstored, MVP returns it but evaluation is deferred â€” static membership onlyâ€” (the spec already half-says this at Â§6 intro).
- Row 7 â†’ Â§6.2: note GET /groups is un-paginated in 1.0.0 (lean shape canonical); keep the paginated form as a documented future-widen.
- Row 5 (?os) â†’ Â§5.2: drop ?os from the 1.0.0 scope set (keep ?deployment_id + time window).

This memo does **not** perform those edits â€” it recommends them pending the operatorâ€™s per-row ruling.

9. anti_bluff_unverified_register

Per Constitution Â§11.4.6, the following are explicitly **not asserted as fact** in this memo:

- **UNVERIFIED â€” effort/risk sizes.** The S/M/L sizes and LOW/MED risk labels are estimates from reading the handlers + store seam, not from a spiked implementation. Absolute hours are not claimed.
- **UNVERIFIED â€” telemetry rich fields availability (row 4).** Whether `store.TelemetryRecord` / the ota-protocol telemetry event already carries `duration_ms` / `bytes_transferred` (and `success` as a first-class field) was **not** confirmed; `handlers_telemetry.go` `TelemetryEventView` does not expose them, and the store record type was not read this pass. If the ingest does not supply them, widening to those fields would produce always-zero columns (a Â§11.4-bluff) â€” hence the â€œphase-2 / deferâ€” qualifier on row 4.
- **UNVERIFIED â€” store-method availability for wider shapes.** The wider audit filter, paginated telemetry query, latest-per-device overview aggregate (`DISTINCT ON`), `member_count` count, and `AddGroupMembers` plural method are named by the spec (`operational_endpoints.md` Â§4.2/Â§5.3/Â§6.5) but their *existence in the pgx/in-memory* `store.Repository` was not verified here. Effort rises if these must be added to the store.
- **UNVERIFIED â€” HelixQA bank path.** `tools/helixqa/banks/atmosphere.yaml` is the constitutional convention (Â§11.4.58); its presence in *this* repo was not confirmed. â€œBankâ€” in Â§4/Â§7 means the projectâ€™s Challenge bank wherever it lives.
- **UNVERIFIED â€” as-built external consumers.** The â€œbreaking wire changeâ€” risk on rows 1/4/8 assumes a consumer that already depends on the lean as-built shape. Whether such a consumer (dashboard, device agent) exists and is pinned to the lean shape was not confirmed; the breaking qualifier is a conservative Â§11.4.6 default (treat as breaking unless proven otherwise), and is exactly why those rows are gated behind an operator Â§11.4.66 decision and a consumer sweep.
- **FACT (confirmed this pass).** Every â€œimpl doesâ€” cell and its `file:func` citation, and every â€œspec wantsâ€” cell, were read directly from the cited source files in `server/internal/api/` and `operational_endpoints.md` during this memoâ€™s preparation (see Â§3).